

in the manifest (such as accessing the device camera) and it tries to access the resource while running, the access is denied by Windows 8. Although AppContainers is enabled by default for Modern UI apps, it can be utilized in desktop apps too by using the API.

win32K.sys

If you know Windows internals well enough, you probably already know about the win32k.sys driver. It's the kernel part of the Windows sub-system. This file is responsible for handling user spaces as well as the primitive (and yet necessary) GDI sub-system. The GDI has almost always been a primary target of attack due to the unprotected API it provides. This makes the win32k.sys a bright attack target as well.

In Windows 8, Microsoft has restricted processes from accessing functions in win32k.sys directly. This means that a significant set of functions which were available for code injection and could later be used to execute malevolent code have been barred. This cranks up overall security of the OS.

However, security is a cat and mouse game. Every time Microsoft comes up with new measures, hackers around the world figure out workarounds. One of the major loopholes of software has been the inability to use the advanced features built into the hardware. Windows 8, for the first time mandates certain hardware features to be installed (PAE, NX, DEP) and utilizes some advanced features in newer, latest processors (such as SMEP in Ivy Bridge). While we consider this to be a step in the right direction, only time will tell if this will keep out hackers.

Enabling Security in Windows 8

Though we interact with computers every day, computers aren't very good at recognizing us. The only way a human can be recognized by a PC is through a 'user account'. This is also where all security restrictions and permissions are applied.

User accounts in Windows 8

We've already mentioned that Windows 8 has a control panel that's split into two – some basic parts lie in the 'PC settings' (Modern UI app) while the more advanced still live in the desktop Control Panel app. User accounts' main features have been sent to the PC Settings app. Since creating a new account, changing password etc. are basic functions of user accounts on a PC, they reside in the Modern UI app rather than in the Control Panel.